

Enhancement of Catfish and Tilapia fisheries in Bangladesh using Microcontroller

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Abstract— With the technological improvements recently the availability of cheap microprocessors and other electrical components has broadened the horizon of its uses in all aspects of life. In this paper we design and implement and actualized a self-monitoring microprocessor based embedded system which will be able to take readings of the parameter from the sensors and give useful information to the user. The paper is focused around Bangladesh and its aquacultural development on Catfish and its family.

Keywords—*Aquaculture, Microprocessor, Tilapia, Catfish, Pangasius, Water-Quality Monitoring, Microcontroller, Bangladesh*

I. INTRODUCTION

Aquaculture in Bangladesh has large scale implications, at an average of **19.71 kg/person/year**. Fish is an important diet staple, providing a rich source of micronutrients and accounting for **60%** of animal protein intake. Research shows that increasing fish consumption and dietary diversity can affect significant developmental changes [1]. Aquaculture also has economical aspects to it and has large impacts on the GDP of Bangladesh supporting millions of aquaculture-farmers. According to the National Department of Fisheries in 2018 fisheries sector has contributed **3.57%** to GDP and **25.30%** to agricultural GDP this numbers are significant for a country with a population of **162 million** [2].

In recent times with increasing demands of fisheries aquaculture has become very popular to meet demands both inside and outside the country. With the

development of microbiological sciences and technology, hybrid fishes such as Tilapia [3] and different species of Catfish [4]. The hybrid fishes, is a cross between two different species that do not mate with each other naturally. Hybrid fish production involves hormone-assisted reproduction. The female channel catfish is given a hormone injection, which induces her to ovulate—release eggs. The fish is anesthetized, and eggs are “stripped,” meaning gently pushed out. Sperm taken from males is mixed onto the eggs, ensuring that the hormone dosages and egg and sperm preparations are right, and dealing with any other issues [5]. There are many researches on hybrid fishes and the modification of the genomes in the species lets the fish to be larger in size compared to the original; the hybrid also let mono-sexual reproduction which allows good offspring count [6]. These hybrids usually thrive in stagnant water and the conditions are suitable for tropical climate of Bangladesh, which makes it profitable for the farmers [7].

With the sheer rise of Bangladesh’s economy in 20th-Century has seen a lot of industrial development as well, but due to the quick shift of economy the waste management of this newly built industries are not up to the standards and affects the surrounding environment heavily [8]. One of the important aspects of our device is to maintain the water quality and optimal conditions for the fishes to grow and reproduce at the maximum possible rate. The design of the device will be such that it is interactive to the changes in the parameters of water and alert the user with the problem and ways to troubleshoot it through.

[1] This paper was developed to produce a cheap system to monitor water quality using Arduino, Raspberry Pi, sensors, android application and a provides great connectivity using Wi-Fi hotspot backbone, the system is also connected with the internet and can be accessed from anywhere. The system has four parameters pH, water color, temperature and electrical conductivity to analyze the water quality.

[2] The paper is focused on constant monitoring of water quality parameters from time to time in order to take preventive measures before actual damage was done, the parameters are Dissolved Oxygen, Ammonia Content, pH, Temperature, Salinity, Alkanity, Nitrate Content. The system also uses Raspberry Pi as the primary microcontroller, the program for collecting sensor data is written in python language and sends data to the cloud database.

[3] This paper is focused on automation of the aquaculture process, it consists of electrical and mechanical components integrated with IoT, the electrical part is consisted of several sensor and a camera which is connected to a Raspberry Pi and a B+ using a web interface. The mechanical parts used for scheduled feeding, mobile manual feeding using the web interface.

[4] The paper is focused on maintaining water quality for drinking purposes, the prototype uses Arduino and are consisted of a pH sensor, Turbidity sensor, Sonar sensor. The system also uses a Wi-Fi module and sends data to “ThingSpeak” a cloud server, and can be used for monitoring the quality of water easily and from anywhere in the world through internet.

[5] This paper is a review on aquaculture monitoring and controlling system using Raspberry Pi and Arduino and sensors used are Temperature sensor, pH sensor, Turbidity sensor, Conductivity sensor. Internet is also integrated in the system and the researchers of the paper has implemented a Neural Network system developed for Aquaculture organism.

LITERATURE REVIEW

No.	Author	PubYear	Brief	Strengths	Obstacles
1	Sajal Saha, Sumaiya Kabir	2018	Uses Raspberry Pi, Arduino and various sensor, the system also uses a smartphone camera and android application to monitor water quality	Gives accurate results for pretty low cost than any other available system.	Two pieces of microcontrollers are used so troubleshooting would be cumbersome.
2	K.Raghu Sita Rama Raju, G.Harish Kumar Varma	2017	Uses Raspberry Pi, the system has the maximum number of sensor modules in the paper, the program is based on Python.	The data collected are very accurate and it uses large number of sensor data and uses analysis for big data.	Too many sensors used, therefore high price and expensive to replace parts.
3	I. S. Akila, Karthikeyan P, Hariharan M. V., Harikrishnan J	2018	Uses Raspberry Pi, various sensors all the components are integrated to IoT using a B+ module. The system also uses automation of feeding using mechanical parts to dispense food.	Cheap and accurate, reduces human effort in fishing.	Mechanical parts are used, moving parts will ware out and is expensive to replace.
4	Fahim Redwan, Shadman Rafid, Ahanaf Hossain Abrar, Bishwajit Banik Pathik	2019	Uses Arduino and few sensors, and are connected to the ThingSpeak cloud server, and can be controlled anywhere from the world using internet.	Very fast continuous transmission of data.	No data on DO level, and high-power consumption due to continuous transmission.
5	Ujwala T. S., Sunnita G. Devareddy, Yamuna S., Vandana S	2020	Uses Arduino, Raspberry Pi, and few sensors and gives analyzed data over the internet and lets automatic and manual control of the mechanism.	Gives accurate data using minimum sensors.	The use of EC sensor used is expensive.

II. PROPOSED HARDWARE & SYSTEM ARCHITECTURE

This section comprises of two sub-sections which are the introduction of required hardware and software technologies and explanation of the functionality of the system.

Arduino Uno, is our proposed microcontroller since all our sensors are compatible with it.

A. Required Hardware and Software

Microcontroller

Arduino Uno is a microcontroller board which is based on the ATmega328P. It has 6 analog input pins and 14 digital input/output pins. Its operating voltage is 5V and recommended input voltage range is 7-12V. Arduino IDE is required to program it. Arduino Uno should be connected to computer with USB cable to be programmed through USB-to-serial converter. It can conduct data communication with computer via serial port [9]

pH Sensor:

pH Sensor, we chose **SKU SEN0161** from Dfrobot (Fig: I). The specifications of the sensor (Table: I) is suitable for our prototype. This sensor needs to be calibrated on factory characteristics and can measure temperature too (Table: II). The probe requires a BNC connector, and has very good accuracy at this budget.

Table: I pH module specification

1	Module Power	5V
2	Module Size	43mm x 32mm
3	Measuring Range	0-14pH
4	Measuring Temperature	0-60°C
5	Accuracy (25 °C)	± 0.1pH
6	Response Time	≤ 1min

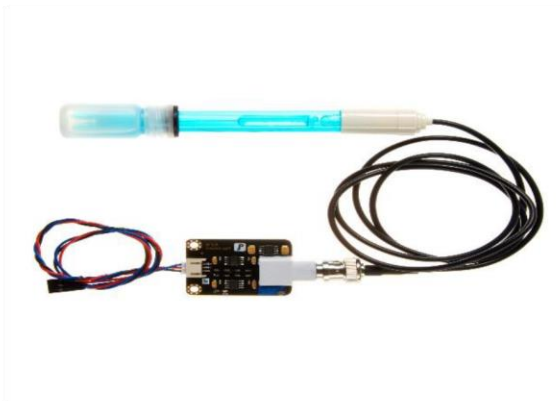


Fig: I pH module

Table: II pH-Voltage Relationship

pH State of Water	pH Level of Water	Voltage at Arduino (mV)
Acidic	0.00	414.12
	1.00	354.96
	2.00	295.80
	3.00	236.64
	4.00	177.48
	5.00	118.32
	6.00	59.16
Neutral	7.00	0.00
Alkaline	8.00	-59.16
	9.00	-118.32
	10.00	-177.48
	11.00	-236.64
	12.00	-295.80
	13.00	-354.96
	14.00	-414.12

Depth and Temperature Sensor:

For Depth and Temperature Sensor, we chose **MS580314BA01-00** from TE Connectivity (Fig: II). The specifications of the sensor (Table: III) is suitable for our prototype, since A high resolution temperature output allows the implementation of a depth measurement systems and thermometer function without any additional sensor. The MS5803- 14BA can be interfaced to any microcontroller. The communication protocol is simple, without the need to programming internal registers in the device. The gel protection and antimagnetic stainless-steel cap protects against 30 bar overpressure water resistant. This sensor needs to be calibrated on factory characteristics. The probe requires a BNC connector, and has very good accuracy at this budget.

Table: III MS580314BA01 Sensor Specification

1	Module Power	1.8V ~ 3.6V
2	Measuring Range	435.11PSI (3000kPa)
3	Operating Temperature	-40°C ~ 85°C
4	Pressure Type	Absolute
5	Output Type	I ² C, SPI



Fig: II MS580314BA01 Sensor

Solar Energy Harvester

A solar photovoltaic cell is a particular semiconductor diode which alters visible light into direct current (DC). The most commonly used cells were made with silicon. Solar photovoltaics directly converts solar energy into electricity. Solar panel absorbs sunlight and transfers the light energy into electricity. A method that causes voltage or electric current in a photovoltaic cell when it is exposed to sunlight is known as the photovoltaic effect [11]. Working on the principle of this effect, when a little amount of silicon is exposed to light, the photovoltaic absorbs photon and release free electrons. This phenomenon is known as the photoelectric effect. The sunlight is converted into direct current (DC) electricity based on the principle of photovoltaic effect [10]

Arduino IDE

Arduino is an open source platform which designs and manufactures single board microcontrollers and microcontroller kits for developing digital devices. Arduino board designs various types of controllers and microprocessors. The boards are integrated with digital and analog input or output pins that may be interfaced on bread boards and other circuits. The boards use serial communications interface which includes universal serial bus (USB) on some models. It can be programmed using C and C++ languages. Based on processing language project integrated development environment (IDE) is used. It aims to provide low cost and easy way for novices [9].

B. System Architecture

Sensors

The proposed water quality monitor, will be able to measure the Dissolved Oxygen (DO), pH, Depth of Water and the temperature in real time. The project has only used two sensors to measure the all the aspects, this makes the prototype cheaper. The DO level in water is not measured using a DO Level Sensor such as SKU:SEN0237-A from DFRobot, which is thrice the cost of the whole project, to make a way through we used the data from temperature sensor to give an approximation of DO. In the paper we have used the relationship between temperature of water and DO using the equation [10]:

$$D = q(0)t + \sum_{m=1}^{\infty} \frac{q^{m+1}(0)t^{m+1}}{(m+1)!} + D_0 \alpha_6$$

$$\text{Where } \alpha_6 = e^{C_0 T \log \theta_2 + \sum_{n=1}^{\infty} C_n}$$

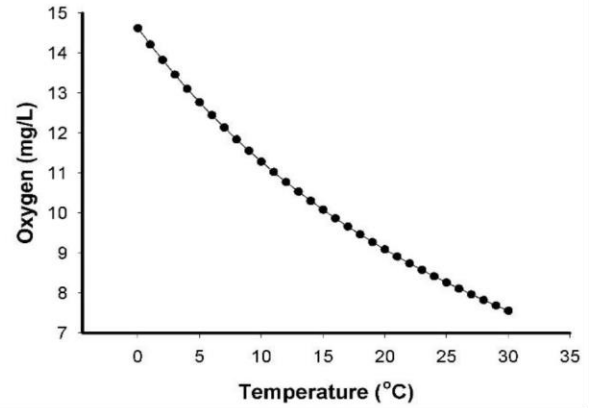
where C is constant of integration

Initially, $t = 0, D = D_0$ & thus

$$C = D_0 e^{C_0 T \log \theta_2 + \sum_{n=1}^{\infty} C_n}$$

This equation approximately calculates the DO, using the velocity of water, bacteria content, rate of aeration etc. Below is the chart for DO vs Temperature graph. (Graph: I)

Graph: I DO vs Temperature Characteristics



The depth sensor we are using is very versatile and can work under high pressure and uses the pressure and temperature to give the depth. The module is calibrated from the factory and hence minimizes the set-up process.

Power Supply

A photovoltaic panel was used to power up the microcontroller and the other sensors. The battery charger is connected to the solar cell. The output of the battery charger is connected to the lithium battery. To convert the voltage from 3.7V to 5V, a step-up voltage booster was connected to the battery (Fig III).

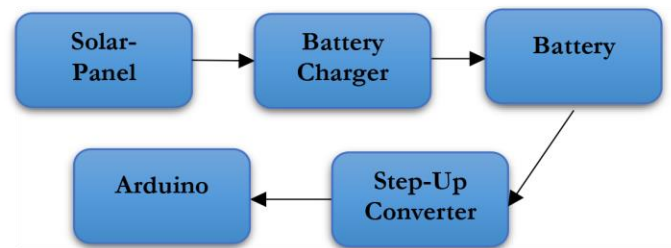


Fig: III Power System Flowchart

For power saving the Microcontroller can be used in sleep mode at night, and a higher capacity battery will provide enough power to last the system a week without any sunlight. Also, the refreshing time can be tweaked down to save power at day time.

II. DESIGN & INTEGRATION

A. System Integration

The data from the sensors are fed in to the Arduino, where the data is calculated and the output of the system is demonstrated using an LCD module. A block diagram representation is below(Fig: IV).

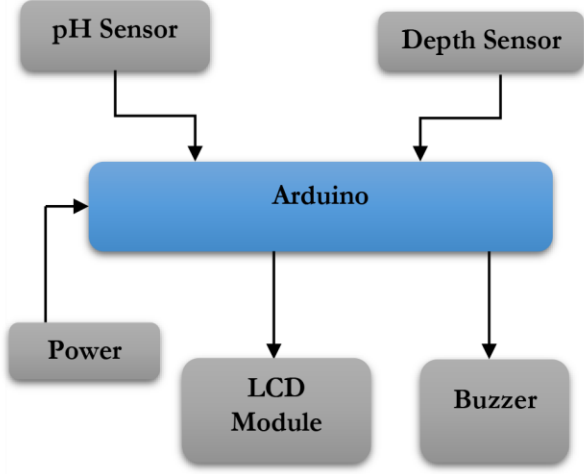


Figure: IV Design Flowchart

B. Water Parameters for Maximum Yield of Tilapia [11]

Various types of water quality parameters for Tilapia was observed during the study period. Water and air temperature ($^{\circ}\text{C}$), transparency (cm), dissolved oxygen (mg/L), pH, ammonia (mg/L), nitrate (mg/L), nitrite (mg/L) were measured after ten days interval. The analysis of variance (ANOVA) was performed to observe whether any difference exists in the mean values of water quality parameters among different treatments.[11] The results in the table below are the optimal values for Tilapia (Table IV).

Table IV: Optimal Values of Parameters for maximum yield

No	Parameters	Optimal Values
1	Water Temperature ($^{\circ}\text{C}$)	31.33 ± 0.22
2	Transparency (cm)	34.38 ± 0.36
3	Dissolved oxygen (mg/L)	5.78 ± 0.16
4	pH	7.38 ± 0.08

After thorough study of the optimal values it can be said that water temperature has not much significance, but since our prototype is meant to be cheap, we used temperature to approximately calculate the value of DO. Hence temperature is a key factor of our design.

C. System Flowchart

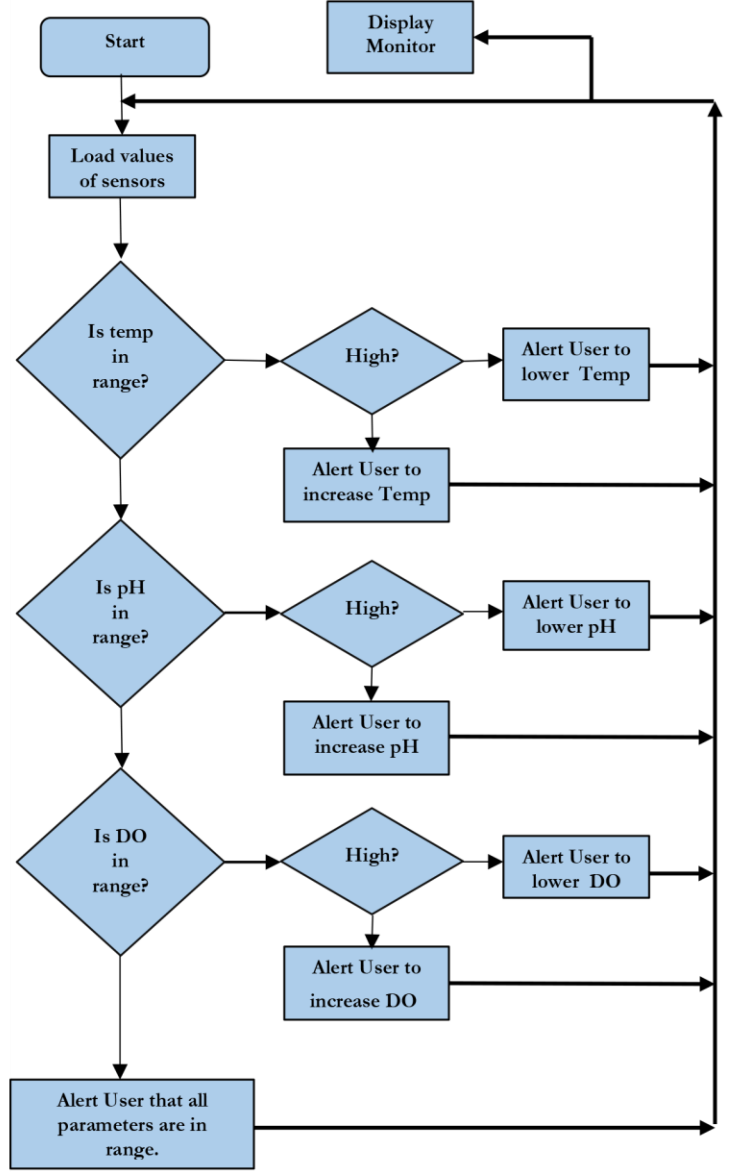


Figure: V System Flowchart for Working Process

The flowchart above (Figure: V) shows the working process of the system, the processor will keep in check that all the parameters are in range. If all the parameters are in between the ranges of the optimal values as in (Table IV) the display module will output the data sensor and the status, if in case any value is out of the optimal range the display will show a hazard status and ring the buzzer. The detailed circuit schematic is in Figure: VI. In the design of the prototype we have added an alarm system integrated to the output and display module, therefore the user can be alerted, a 60dB Buzzer can be heard from 20m away or more.

D. Circuit Schematic

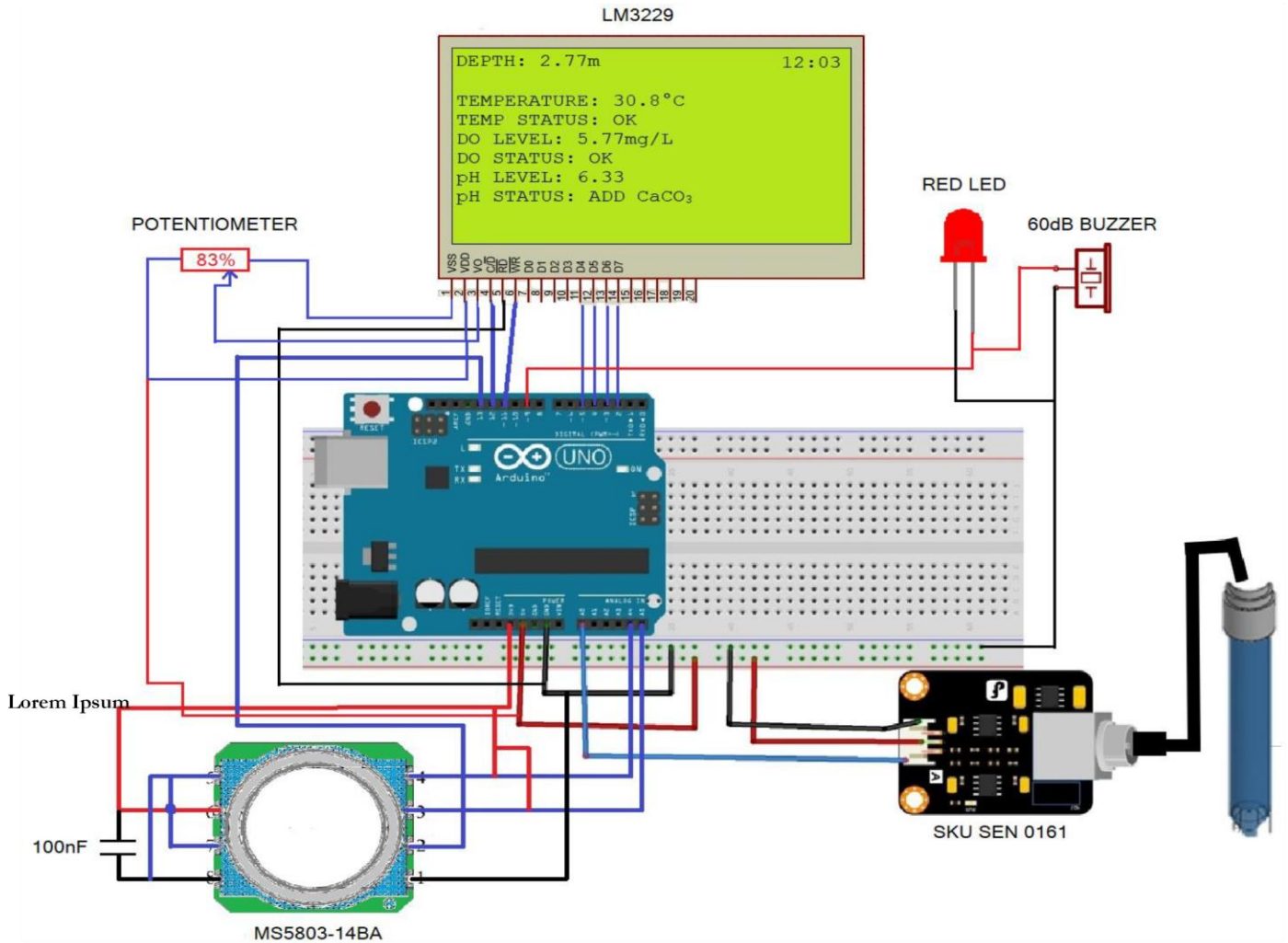


Figure: 6 Graphical Circuit Schematics

III. FUTURE WORK

The system to monitor the water quality for aquaculture (SMWQA) lets constant monitoring of water with proper transitional sleep modes to save power, the project can be

further optimized by using salinity, ammonia level and Electrical Conductivity of the water, since this play an important role in the growth of Tilapia's. In the research [11] Air temperature (°C): Air temperature was same for all the experimental ponds on each sampling day. The system can be integrated with internet to monitor it ubiquitously from any location and be used in a Neural Network Backbone. The system can be also be connected to the Satellite to constantly assess environmental changes in the vicinity **aquaculture** sites by, for example, detecting changes in sea grass beds, coral reefs, mangrove forests, erosion or deposition on inter-tidal mud flats.

The first table below shows the trials for the certain parameters, the results for growth parameters for the trials. We have used the parameters for Trial 2 since that has maximum growth.

Table V(a.) Parameters to check optimal conditions

Parameters	Trial 1	Trial 2	Trial 3	Sgfn
Air Temperature (°C)	30.25±0.14	30.25±0.14	30.25±0.14	NS
Water Temperature (°C)	31.27±0.22	31.33±0.22	31.20±0.21	NS
Transparency (cm)	36.63±0.36	34.38±0.36	34.98±0.36	S
Do (mg/L)	4.56±0.15	5.78±0.16	4.62±0.15	S
pH	6.84±0.03	7.38±0.08	6.87±0.04	S
Ammonia (mg L-1)	0.31±0.02	0.40±0.03	0.80±0.04	S
Nitrate (mg L-1)	0.02±0.0	0.03±0.0	0.13±0.4	NS

Table V(b.) Parameters of Optimal Condition for Maximum Yield

Growth-Parameters	Trial 1	Trial 2	Trial 3	Sgfn
Initial weight (g)	6.0±0.00	6.0±0.00	6.0±0.00	NS
Final weight (g)	197±1.00	219±1.00	212±2.00	S
Weight Gain (g)	191±1.00	213±1.00	206±2.01	S
% of weight gain	3183.33±16.72	3550±16.72	3433.33±3343	S
Survival (%)	85.5±0.50	91.75±0.75	85.25±1.25	S
Production (kg/120 days)	33.69±26.08	40.19±513.53	36.14±189.56	S

IV. CONCLUSION

Our prototype uses only two sensors, therefore the temperature measured using the **MS580314BA01-00**, the reading of temperature is not much accurate since, the sensor is designed to measure altitude and depth. But to achieve an Embedded System for SMWQA at such a low-cost limit us using other sensors also using more sensors might limit the use of Arduino Uno and use Arduino Mega which will have more pins.

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